

NAME: SOLUTIONS

UIN:

Math 553 — Spring 2026— Midterm 2

Total points 100.

Questions: 5

Explain all answers. If you cannot solve a problem, at least explain the relevant theorems or methods.

Duration: 1 hour (you may stay for up to 2 hours)

DO NOT WRITE ON THE BACK OF THE PAGE.

Work on the backs of pages will not be scanned or graded.

Write extra work on the last two pages (blank).

$B(x, r)$ is the ball centered at x with radius r .

Books and notes may be used (anything on paper).

No communication or consultation with any person or electronic resource about the exam.

No laptops, electronic devices or phones, except that you may consult books and notes you have stored on these devices. Web searching and LLMs are not permitted.

Open note files on your devices BEFORE the start of the exam. No typing during the exam.

Honor code statement

I did not use electronic devices, except to consult books and notes on my devices. I did not communicate or consult with any person or electronic resource about the contents of this exam.

Signature:

Date: Thursday April 16, 2026

Problem 1. (20=15+5 points)

(a) Suppose $u(x, t)$ is smooth for $x \in \mathbb{R}$ and $t \geq 0$, with u having compact support as a function of x at each time t . Write $u(x, 0) = f(x)$ and assume u solves Burgers' equation with viscosity, that is, $a > 0$ and

$$u_t + uu_x = au_{xx}.$$

Show that the energy $E(t) = \frac{1}{2} \int_{-\infty}^{\infty} u(x, t)^2 dx$ is dissipated.

Solution:

$$\begin{aligned} E'(t) &= \int_{-\infty}^{\infty} uu_t dx && \text{by differentiating through the integral wrt } t \\ &= \int_{-\infty}^{\infty} u(au_{xx} - uu_x) dx && \text{by the PDE} \\ &= a \int_{-\infty}^{\infty} uu_{xx} dx - \frac{1}{3} \int_{-\infty}^{\infty} (u^3)_x dx \\ &= -a \int_{-\infty}^{\infty} u_x^2 dx \leq 0 && \text{by parts and the fundamental theorem,} \end{aligned}$$

using that $u(x, t)$ equals zero for all large $|x|$.

(b) "If the solution is initially small then it will remain small." Express this idea precisely, and prove it.

Solution:

Precise statement, using the L^2 sense of smallness:

Let $\varepsilon > 0$. If $\int_{-\infty}^{\infty} f(x)^2 dx < \varepsilon$ then $\int_{-\infty}^{\infty} u(x, t)^2 dx < \varepsilon$ for each $t > 0$.

Proof by dissipation of energy: $\frac{1}{2} \int_{-\infty}^{\infty} u(x, t)^2 dx = E(t) \leq E(0) = \frac{1}{2} \int_{-\infty}^{\infty} f(x)^2 dx < \varepsilon$.

(c) (*Optional, no credit.*) Why does our usual technique for proving uniqueness not work on this PDE?

Solution:

Suppose u_1 and u_2 are two solutions of the PDE. Our usual technique for proving uniqueness is to consider the difference $w = u_1 - u_2$ and prove that w satisfies the same PDE, and then show by an energy method (or by a maximum principle) that $w \equiv 0$. But in this problem, the nonlinear term in the PDE prevents us from getting a PDE for w . Subtracting the equations for u_1 and u_2 gives that

$$w_t + (u_1 u_{1,x} - u_2 u_{2,x}) = aw_{xx}$$

and we cannot express the middle terms using w .

Problem 2. (15=10+3+2 points) (a) Let $b > 0$. Show that

$$\begin{aligned}u_t &= u_{xx} + bu, & x \in \mathbb{R}, \ t > 0, \\u(x, 0) &= \delta_0(x),\end{aligned}$$

has solution $u(x, t) = e^{bt-x^2/4t}/\sqrt{4\pi t}$.

Solution: By HW 7 #5, we know that $v(x, t) = e^{-bt}u(x, t)$ satisfies the heat equation. Its initial condition in this problem is $v(x, 0) = u(x, 0) = \delta_0(x)$. So a solution for v is the fundamental solution of the heat equation in 1 dimension, that is, $v(x, t) = e^{-x^2/4t}/\sqrt{4\pi t}$. Hence

$$u(x, t) = e^{bt}v(x, t) = \frac{1}{\sqrt{4\pi t}}e^{bt-x^2/4t}$$

(b) Fix $c > 0$. The point $x(t)$ at which u achieves the value c (meaning $u(x(t), t) = c$) is

$$x(t) = \sqrt{4bt^2 - 2t \log t - 4t \log(c\sqrt{4\pi})}.$$

Evaluate $\lim_{t \rightarrow \infty} \frac{x(t)}{t}$.

Solution:

$$\frac{x(t)}{t} = \sqrt{4b - 2\frac{\log t}{t} - \frac{4}{t} \log(c\sqrt{4\pi})} \rightarrow \sqrt{4b} \quad \text{as } t \rightarrow \infty.$$

Interestingly, this limiting value $\sqrt{4b}$ does not depend on c .

(c) The answer to part (b) is called the **spread rate** of the PDE. Explain intuitively what b represents in the PDE and hence why the spread rate should get larger when b gets larger.

Solution:

The constant b represents a rate of heat generation (or of population growth). When that rate gets larger the solution gets larger, and so the location $x(t)$ where u takes value c should be farther away from the origin than previously, which means the solution is “spreading” faster.

Problem 3. (20=14+6 points) Let $R > 0$ and $x = (x_1, x_2) \in \mathbb{R}^2$.

(a) Suppose $u(x)$ is a C^2 -smooth function on the plane $\mathbb{R}^2$. Show that

$$\int_{B(0,R)} (R^2 - |x|^2) \Delta u \, dx = -4 \int_{B(0,R)} u \, dx + 2R \int_{\partial B(0,R)} u \, dS.$$

Solution:

The left side has Δu and the right side has u . So we apply (Green2) to $u(x)$ and to the function

$$v(x) = R^2 - |x|^2 = R^2 - x_1^2 - x_2^2,$$

obtaining that

$$\begin{aligned} \int_{B(0,R)} (R^2 - |x|^2) \Delta u \, dx &= \int_{B(0,R)} u \Delta v \, dx + \int_{\partial B(0,R)} \left(v \frac{\partial u}{\partial \nu} - \frac{\partial v}{\partial \nu} u \right) dS \\ &= -4 \int_{B(0,R)} u \, dx + \int_{\partial B(0,R)} (0 - (-2R)u) \, dS \end{aligned}$$

since $\Delta v = -4$ and on $\partial B(0, R)$ we have $v = 0$ and $\partial v / \partial \nu = \partial v / \partial r = -2|x| = -2R$.

Comment: (Green1) does not help on this problem, because we do not want ∇u in our formulas.

(b) Use part (a) to show that if u is subharmonic then the mean value over the disk of radius R is less than or equal to the mean value over the circle. That is, show

$$\int_{B(0,R)} u \, dx \leq \int_{\partial B(0,R)} u \, dS.$$

Solution:

(b) Since $\Delta u \geq 0$ by subharmonicity, and $R^2 - |x|^2 \geq 0$ in $B(0, R)$, we deduce from part (a) that

$$4 \int_{B(0,R)} u \, dx \leq 2R \int_{\partial B(0,R)} u \, dS.$$

Dividing both sides by $4\pi R^2$ yields the desired inequality on the mean values over the disk and circle.

Problem 4. (20=10+10 points)

(a) State the Mean Value Theorem characterizing harmonic functions, on a domain $\Omega \subset \mathbb{R}^n$.

Solution:

The Mean Value Theorem as stated in class says that for $u \in C^2(\Omega)$:

$$\begin{aligned} & u \text{ is harmonic } (\Delta u = 0) \\ \iff & u(x) = \oint_{\partial B(x,r)} u \, dS && \text{for all } x, r \text{ with } \overline{B(x,r)} \subset \Omega \\ \iff & u(x) = \oint_{B(x,r)} u(y) \, dy && \text{for all } x, r \text{ with } \overline{B(x,r)} \subset \Omega \end{aligned}$$

(b) Suppose $u(x)$ is harmonic on $\mathbb{R}^n$. Let O be an $n \times n$ orthogonal matrix (so that O represents a rotation and/or reflection on $\mathbb{R}^n$). Show the function $v(x) = u(Ox)$ is harmonic, **by using the Mean Value Theorem**.

Solution:

Let $x_* \in \mathbb{R}^n$ and $r > 0$. We have

$$\begin{aligned} & \frac{1}{|B(x_*, r)|} \int_{B(x_*, r)} v(x) \, dx \\ &= \frac{1}{|B(x_*, r)|} \int_{B(x_*, r)} u(Ox) \, dx \\ &= \frac{1}{|B(Ox_*, r)|} \int_{B(Ox_*, r)} u(y) \, dy && \text{by letting } y = Ox, dy = dx \\ &= u(Ox_*) && \text{by the Mean Value Theorem for } u \\ &= v(x_*). \end{aligned}$$

Thus v satisfies the mean value property, and so by part (a), v is harmonic.

Problem 5. (25=5+5+15 points) Let f be a smooth function with compact support in 3-dim space $\mathbb{R}^3$.

(a) Write down the fundamental solution $K(x)$ for the Laplacian, solving $-\Delta K = \delta_0$ in $\mathbb{R}^3$.

Solution: $K(x) = \frac{1}{4\pi|x|}$

(b) Write down the solution of Poisson's equation $-\Delta u = f$ in $\mathbb{R}^3$ with $u(x) \rightarrow 0$ as $|x| \rightarrow \infty$.

Solution: $u(x) = \int_{\mathbb{R}^3} K(x-y)f(y) dy$

(c) Assume f is radially symmetric, meaning $f(y) = F(r)$ when $y = r\xi$, and suppose $F(r) = 0$ for all $r \geq 1$. Assume $|x| > 1$. Show that $u(x) = mK(x)$, where $m = \int_{\mathbb{R}^3} f(y) dy$. *Hints.* Spherical coordinates in the integral defining $u(x)$. If you use a mean value formula from #4(a), draw a diagram to verify the hypotheses.

Solution:

$$\begin{aligned} u(x) &= \int_{\mathbb{R}^3} K(x-y)f(y) dy \\ &= \int_0^\infty \int_{\mathbb{S}^2} K(x-r\xi)f(r\xi) d\xi r^2 dr && \text{by spherical coordinates} \\ &= \int_0^1 \int_{\mathbb{S}^2} K(x-r\xi) d\xi F(r)r^2 dr && \text{since } f = F \text{ is radial and equals 0 for } r \geq 1 \\ &= \int_0^1 4\pi K(x) F(r)r^2 dr && \text{by Mean Value Theorem applied to the harmonic function } K, \text{ on } \partial B(x,r), \\ &= K(x) \int_0^1 4\pi F(r)r^2 dr = K(x) m && \text{by spherical coordinates to evaluate } m. \end{aligned}$$

To justify the use of the Mean Value Theorem, notice that K is harmonic everywhere except the origin, and the closed ball $\overline{B(x,r)}$ does not contain the origin. (Draw a diagram, noting that $|x| > 1$ and $r < 1$.)

Comment. Part (c) means physically that the gravitational potential of a radially symmetric planet has the same value (outside the planet) as if all the mass m were concentrated at a point mass in the center.